REVIEW
Medical Data Mining 2026;9(2):11. https://doi.org/10.53388/MDM202609011

AI-based learning models for the life cycle prediction and detection of diabetes disorders: A comprehensive perspective

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Competing interests
The authors declare no conflicts of interest.

Acknowledgments
This research received no external funding.

Abbreviations
ACVO, adaptive chaotic variable optimization; AdaBoost, Adaptive Boosting; AI, artificial intelligence; ANN, artificial neural network; ANOVA, analysis of variance; BCQ, batch-constrained Q-learning; BEC, bagging ensemble classifier; CART, classification and regression tree; CGM, continuous glucose monitoring; CNN, convolutional neural network; DCNN, deep convolution neural networks; DL, deep learning; DNNs, deep neural networks; EHRs, electronic health records; ENN, edited nearest neighbour; KNN, K-nearest neighbours; LASSO, Least Absolute Shrinkage and Selection Operator; LSTM, long short-term memory; ML, machine learning; MRMR, minimum redundancy maximum relevance; PCA, principal component analysis; QA, quality assessment; RFE, recursive feature elimination; RL, reinforcement learning; RQs, research questions; SHAP, Shapley Additive exPlanations; SLR, systematic literature review; SMOTE, synthetic minority oversampling technique; SVM, support vector machine; t-SNE, t-distributed stochastic neighbour embedding.

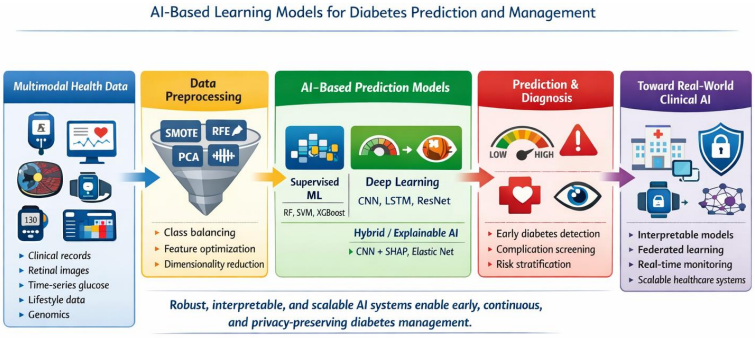
Citation
Nazim M, Ansari MA, Ahmad S, Arif M. AI-based learning models for the life cycle prediction and detection of diabetes disorders: A comprehensive perspective. *Med Data Min.* 2026;9(2):11. doi: 10.53388/MDM202609011.

Peer review information
Medical Data Mining thanks all anonymous reviewers for their contribution to the peer review of this paper.
Executive editor: Ming-Hao Wang.
Received: 12 June 2025; **Revised:** 04 July 2025; **Accepted:** 10 April 2026; **Available online:** 15 April 2026.
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Abstract

This paper aims to conduct a systematic literature review (SLR) using an artificial intelligence (AI) approach to predict and diagnose diabetes mellitus. After reviewing the literature published from 2015–2025, the paper aims to identify the most effective AI techniques, the most used datasets, the most widely used data preprocessing techniques, and the most common issues. After analyzing the literature, it has been found that convolutional neural networks (CNNs) and long short-term memory (LSTM) networks are deep learning models that have shown high accuracy in diabetes prediction. Recursive feature elimination (RFE) and SMOTE are feature selection techniques that have significantly improved model accuracy, training time, and interpretability. Amidst this technological advancement, some existing issues persist: data imbalance, the inapplicability of techniques, computational limitations, and a lack of real-time application in a healthcare environment. The literature review has also identified the need for robust, interpretable, and scalable AI systems capable of handling large volumes of data, including real-world data, in the healthcare industry. Furthermore, it has been identified that the benefits should be integrated with wearable health monitoring systems and the development of privacy-preserving models to ensure continuous, secure, and proactive diabetes management.

Keywords: artificial intelligence; machine learning; diabetes prediction; deep learning models; healthcare data analytics



Introduction

Diabetes mellitus is associated with a chronic metabolic disorder that affects a massive number of people around the world. The rising incidence of diabetes is alarming; hence, the cost of its management in terms of health and the economy is high [1]. According to the International Diabetes Federation, in the year 2021, as many as 537 million adults worldwide suffered from diabetes, and the number of diabetic patients is expected to rise alarmingly by the year 2045 [2]. Among the most serious potential complications arising from diabetes are cardiovascular disease, chronic kidney disease, and diabetic neuropathy. However, thanks to the rising availability of health corporate data and information, along with advanced artificial intelligence (AI) and machine learning (ML) models, the conventional management of diabetes is revolutionized by assisting in the early detection and active management approaches for diabetes. With the help of AI, new approaches such as deep learning (DL) and improved ensemble models have enabled significantly greater accuracy in predicting diabetes than conventional approaches [3].

The most powerful potential of AI lies in prediction and type 2 diabetes management models using intelligent learning, which provide more advanced treatment through accurate diagnosis and identification of high-risk patients for monitored, better-tailored treatment approaches [4]. There has been much development and use of AI and ML algorithms for diagnosing various medical conditions, including diabetes. Although there has been some progress, there are still obstacles that form a challenge for managing diabetes. Most current diagnostic methods include clinical diagnosis and laboratory testing, which may not be inexpensive, rapid, or available in resource-poor nations [5].

Effective, inexpensive diagnostic tools are essential to the medical field. This has led to advances in research on AI models [6]. Although the predictive targets of ML and DL models have shown promise in predicting diabetes, there are still some gaps. Many current models are trained on small datasets, leading to overfitting and making them ineffective in real-world medical settings. Moreover, most AI models suffer from generalization problems [7].

A model developed and trained on one set of people may not perform well when applied to another set because of differences in their genetics, lifestyles, and even socioeconomic status. One more significant reason that drives this research is the use of AI in predictive modeling for proactive care. By using AI for diabetes prediction, healthcare professionals can identify early symptoms of the disease and take appropriate action before it becomes a life-threatening situation [8]. Moreover, by combining AI with wearable health monitoring devices, glucose levels are provided in real time, enabling appropriate action for high-risk patients. This study systematically reviews the literature on AI-based prediction and detection of diabetes. It focuses on supervised and unsupervised learning, DL, reinforcement learning, and hybrid models. Moreover, the analysis of this study centers on the key datasets that are employed, the techniques of implementation sampling, and the hurdles in the usage of AI for predicting diabetes.

This research proposes to outline a systematic approach in the area that integrates results from existing research. In addition, this research proposes applying AI-based models for predicting the onset of diabetes in the clinical environment while improving their interpretability, fairness, and real-time applicability. The application of AI in diabetes care and its potential to reduce healthcare costs while improving patient satisfaction and service are also discussed. Although AI methods for diabetes prediction appear to have tremendous potential, they also have some limitations. Most AI models are based on publicly available, but trivial, data sets that do not reflect the real world because they have small populations and are not diverse. The lack of transparency in decision-making processes in DL models (commonly known as the black-box problem) is a barrier to primary care physicians accepting AI decision-support models, hindering the application of AI technology in the clinical environment. In addition, most other AI models are computationally intensive, making them difficult to apply. Most of them provide a paradigm shift in continuous glucose monitoring (CGM) and wearable sensor data models linking tremors to capture changes in patient activity and enhance monitoring intensity, which, combined with imprecise models, reduces efficiency. Deceptive information is another issue, as datasets that are not balanced often lead to biases in model training, thereby influencing the diagnostic accuracy of those whose populations are not well-represented. Moreover, AI models require a huge amount of patient data, which raises issues of privacy and security and poses ethical concerns regarding data protection for AI application in the healthcare sector. To address the needs mentioned above, the following are considered.

- Performing SLRs on AI diabetes prediction models regarding their effectiveness and challenges.
- Discussion on the collection of commonly used datasets in diabetes prediction and their impact on model performance.
- Analysis of preprocessing AI model data for enhanced accuracy in AI predictive models.
- Clarification on the weaknesses and improvability of AI diagnosis of diabetes.
- Discussion on the emerging recommendations for improved AI diabetes managed healthcare systems.
- Emphasis on continuous monitoring of diabetes through real-time AI connectivity services with wearable healthcare devices.

Taking into consideration the emergent trend in the use of AI for predicting diabetes, a systematic literature review is required to provide a synopsis of the literature and then highlight the gaps, a consolidated knowledge of the available knowledge and trends in this field, thus, a comparatively new and multidimensional perspective for researchers and practitioners.

The paper is divided as follows: In Methods section, the research methodology is described, including the research questions, search strategy, and criteria for selecting studies. A general overview of the results, including the classification of results by AI methods, datasets, preprocessing techniques, and diabetes prediction challenges, is presented in Results section. In Discussion section, the discussion related to the topic is presented. Conclusion section concludes the study by highlighting key aspects and providing directions for further research to enhance AI-based diabetes predictive models.

Methods

Kitchenham's guidelines have been adopted as the research methodology to explore key issues in the domain of AI-driven learning models for the detection of diabetes [9, 10]. The following essential steps are involved in this process: formulating research questions (RQs), developing a robust search strategy, selecting relevant studies, and synthesizing the collected data.

Research questions

The goal of this SLR is to uncover research gaps in the field of AI-based detection or diagnosis of diabetes. To address this objective, the following research questions have been carefully developed as part of the SLR process:

Which of the AI techniques are most used for the detection or prediction of diabetes?

What datasets are commonly used to train and validate the AI models for diabetes prediction?

Which data pre-processing techniques are most effective in enhancing the performance of AI models used for diabetes prediction?

What are the key limitations of the present AI-based diabetes prediction methods, and how might these limitations be addressed?

Search strategy

A search string has been formulated using the key terms from the research questions to extract primary studies from the following eight

electronic databases: ACM Digital Library, Google Scholar, IEEE *Xplore*, ResearchGate, ScienceDirect, Semantic Scholar, Springer, and Taylor & Francis.

Search string: *((AI OR artificial intelligence OR machine learning OR deep learning OR ensemble learning OR supervised learning OR unsupervised learning) AND (method OR approach OR technique OR framework OR model OR architecture OR algorithm) AND (detection OR prediction OR diagnosis OR identification OR classification) AND (diabetes OR diabetes mellitus OR type 1 diabetes OR type 2 diabetes OR gestational diabetes))*

Study selection process

In this work, the data have been extracted from the studies published during the last eleven years (i.e., from 2015 to 2025). The complete selection process of the studies is demonstrated in Figure 1. In the initial search process, 446 studies were identified from the eight different digital database resources as: 17 studies on ACM Digital Library, 39 studies on Google Scholar, 119 studies on IEEE *Xplore*, 10 studies on ResearchGate, 83 studies on ScienceDirect, 76 studies on Semantic Scholar, 96 studies on Springer, and 6 studies on Taylor & Francis.

The selected studies were scrutinized based on the titles of the studies to remove duplicates and irrelevant studies. As a result, 387 studies have been selected as the primary studies. Of these primary studies, 243 were selected based on their abstracts and conclusions. We reviewed the references of the selected 243 studies to ensure that no relevant studies were missed in the initial phase of searching. Consequently, we found 19 relevant studies from the references of 243 selected studies. Subsequently, 262 were used for the quality assessment (QA) process. The QA process was performed based on the following assessment questions to ensure the quality and the relevancy of the selected studies: (QA1)-"*Is the sample size adequate to draw reliable conclusions?*", (QA2)-"*Are the methods used well described and appropriate?*", (QA3)-"*Are the results clearly presented and statistically analyzed?*", (QA4)-"*Are the limitations of the current methods discussed?*", (QA5)-"*Are there any innovative AI-driven solutions presented for diabetes management? If yes, does the study discuss the practical implementation of these AI-driven solutions?*". The following scores were used for the QA: (a) Yes = 1, (b) Partly = 0.5, and (c) No = 0.

For example, in study by Shahin et al. [11], a large and adequate dataset is used; therefore, QA-1 is rated as 1.0. QA-2 is scored 1.0 because the methods are well described in this study. The results are presented and provide a suitable statistical analysis; therefore, a score of 1.0 is given to QA-3. However, this study did not extensively discuss the methods' limitations; therefore, a QA-4 is rated at 0.5. QA-5 scored 1.0 because of the innovative application of restricted Boltzmann machines in this domain. Similarly, the QA score of the remaining studies is computed. For further analysis, we only selected the studies with a score value greater than 4.0. Consequently, 127 studies have been selected to address the RQs. The complete list of 127 selected studies and their quality score values is presented in Supplementary Table S1 and Supplementary Table S2, respectively.

Data synthesis

This study's data related to RQ-1, RQ-2, RQ-3, and RQ-4 are extracted from the 127 studies and illustrated using bar charts, tables, and text. The data related to RQ-1, i.e., the year-wise distribution of the studies, is presented using the bar chart. Tables are used to present the results of RQ-2 and RQ-3, whereas the result of RQ-4 is given in the form of text.

Threats to validity

We created a search string to find out the primary studies from eight different digital libraries. Practically, extracting all studies from the terms used in the RQs is impossible. To counter this threat, a manual scrutiny of the studies was conducted to identify the relevant primary studies that might have been overlooked in the initial search phase of the SLR. The quality assessment of the studies was carried out independently to reduce the inaccuracy of the retrieved data.

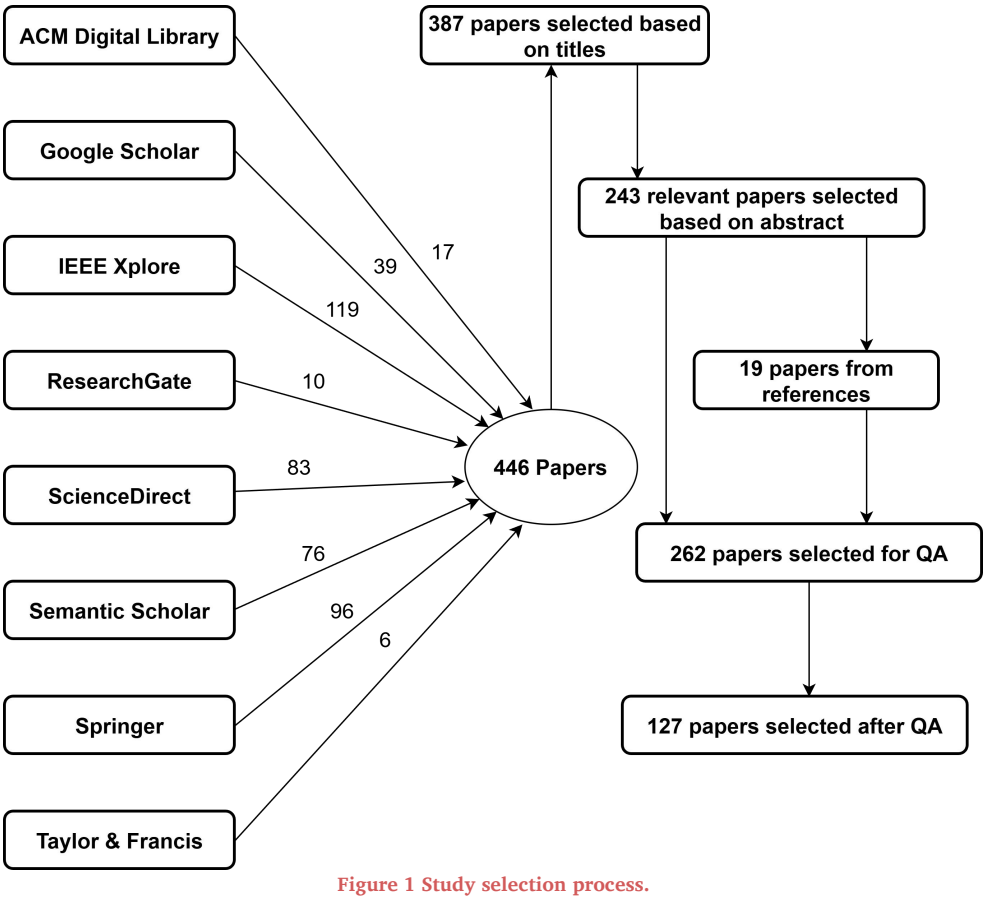


Figure 1 Study selection process.

Results

Figure 2 shows how many published studies concerning AI-based diabetes studies have been published each year between 2015 and 2025. Clearly, the trend is upward as there is a gradual rise throughout the years, with 2 studies in 2015 increasing to 35 studies in 2023. There are marginal decreases in 2024 (25 studies) and also in 2025 (11 studies till now), which can be explained by the fact that 2025 is not covered completely yet. All in all, the statistics indicate that there is a rising interest of the research community in this field, especially starting to increase more rapidly in 2020.

The analysis of the RQs is shown in Supplementary Table S2, using the set of 127 high-quality papers after a rigorous screening and quality evaluation exercise. These articles were identified systematically using title screening, abstract and reference mining on eight major digital libraries. The chosen articles establish the basis upon which the research questions are answered, giving all-around information on AI-based methods, sources of data, and information preprocessing involved in the prediction and diagnosis of diabetes.

RQ-1: Which of the AI techniques are most used for the detection or prediction of diabetes?

Mathew and Sathyalakshmi used the adaptive chaotic variable optimization (ACVO) technique that works well for non-linear optimization problems [12]. To increase the overall performance and robustness by iteratively modifying the weights of weak classifiers and emphasizing the misclassified instances in the subsequent iterations, an ensemble learning technique named Adaptive Boosting (AdaBoost) Algorithm was used [13–17]. Islam et al. applied AdaBoostM1, Bootstrap Aggregating (Bagging) Meta-Algorithm, J48, and One Rule (OneR) [18]. AdaBoostM1 is an advanced version of the AdaBoost algorithm. It is specifically designed for multi-class classification tasks, using a meta-learning technique to combine weak hypotheses and maximize prediction accuracy across various datasets. Bagging is an ensemble learning method used to train multiple models on different bootstrap samples of the data and aggregate their predictions to improve stability and accuracy. J48 is a decision tree algorithm. It implements the C4.5 algorithm and robustly handles missing values and continuous attributes. OneR is a simple rule-based classifier. It generates a single rule for the classification by selecting features with the minimum error rate. Alhassan et al. implemented various techniques like Gated-Recurrent Unit, Adaptive Neuro-Fuzzy Inference System, and principal component analysis (PCA) [19]. They also used the Temporal Predictive DL Models to leverage temporal dependencies in time-series data. This technique combines the learning powers of neural networks with the interpretability of fuzzy logic to provide a strong framework for modelling complex and nonlinear systems, especially when decision-making under uncertainty is required.

Lakshmi and Sargunam used AlexNet, DenseNet121, Inception v3, ResNet50, and VGG16 [20]. AlexNet is a deep convolutional neural network architecture. It introduced innovative features like ReLU activation and dropout. It significantly advances state-of-the-art image classification and serves as a foundation for subsequent DL models. DenseNet121 is an advanced version of DenseNet. This method leverages 121 layers to model complex data distributions with fewer computations. Inception v3 is a DL model. It was used for image recognition. ResNet50 is a variant of ResNet with 50 layers. It balances depth and computational efficiency for high-performance visual tasks. VGG16 is another very famous deep convolutional neural network with 16 layers. It is known for its uniform architecture and effectiveness in large-scale visual recognition tasks. A computational model inspired by the human brain and termed artificial neural network (ANN) was used [21–25]. It consists of interconnected nodes termed neurons. Its capability of processing and learning patterns in the data through adaptive weight adjustments makes it effective for solving complex classification and regression problems. Guleria et al. applied two methods, analysis of variance (ANOVA) and minimum redundancy maximum relevance (MRMR) [26]. ANOVA was used to analyze variations among group means, evaluating the impact of one or more factors by comparing the variance within groups to the variance between the groups. In contrast, MRMR is a feature selection method used to select features that are highly relevant to the target variable and have minimal redundancy. Dey et al. used an optimization technique formed by combining the backpropagation approach for training neural networks with genetic algorithms [23]. In our SLR, we found various AI-based techniques used to detect or predict diabetes. The different techniques based on their learning types are given in Table 1.

Table 1 Learning type-based categorization of AI techniques used for the diabetes prediction

Learning Type	Methods	References
Supervised Learning	Decision Tree, Random Forest, Logistic Regression, Support Vector Machine, KNN, ANN, Naive Bayes, XGBoost, AdaBoost, Gradient Boosting, LASSO, Linear Regression, Multivariate Logistic Regression, classification and regression tree	[13–16, 21–25, 27–55]
Unsupervised Learning	K-means Clustering, t-distributed stochastic neighbour embedding, PCA, Modified Fuzzy Clustering, Isolation Forest	[31, 47, 48, 56–59]
Reinforcement Learning	Batch-Constrained Q-learning, Offline Reinforcement Learning	[60]
Semi-Supervised Learning	SMOTEENN Hybrid Sampling Technique, CNN-SVD Hybrid Model	[61, 62]
Deep Learning	CNN, RNN-LSTM, GRNN, Deep Belief Network, Deep CNN, ResNet, DenseNet, VGG (11, 16, 19), LSTM	[11, 20, 22, 23, 28, 40, 47, 48, 53, 63–69]
Meta-Learning	Bagging, AdaBoostM1, Ensemble Soft Voting Classifiers	[18, 70]
Feature Engineering	RFE, Shapley Additive Explanations, Local Interpretable Model-Agnostic Explanations	[32, 51, 69, 71]
Optimization Techniques	Backpropagation Genetic Algorithm, Exponentially Weighted Moving Average, ACVO	[12, 23, 51]

ACVO, Anti-Coronavirus Optimization; AdaBoost, Adaptive Boosting; AdaBoostM1, AdaBoost M1; ANN, artificial neural network; CNN, convolutional neural network; DenseNet, Densely Connected Convolutional Network; GRNN, Generalized Regression Neural Network; KNN, K-nearest neighbours; LASSO, Least Absolute Shrinkage and Selection Operator; LSTM, long short-term memory; PCA, principal component analysis; ResNet, Residual Network; RFE, recursive feature elimination; RNN-LSTM, Recurrent Neural Network-Long Short-Term Memory; SMOTEENN, SMOTE combined with Edited Nearest Neighbours; VGG, Visual Geometry Group; XGBoost, eXtreme Gradient Boosting.

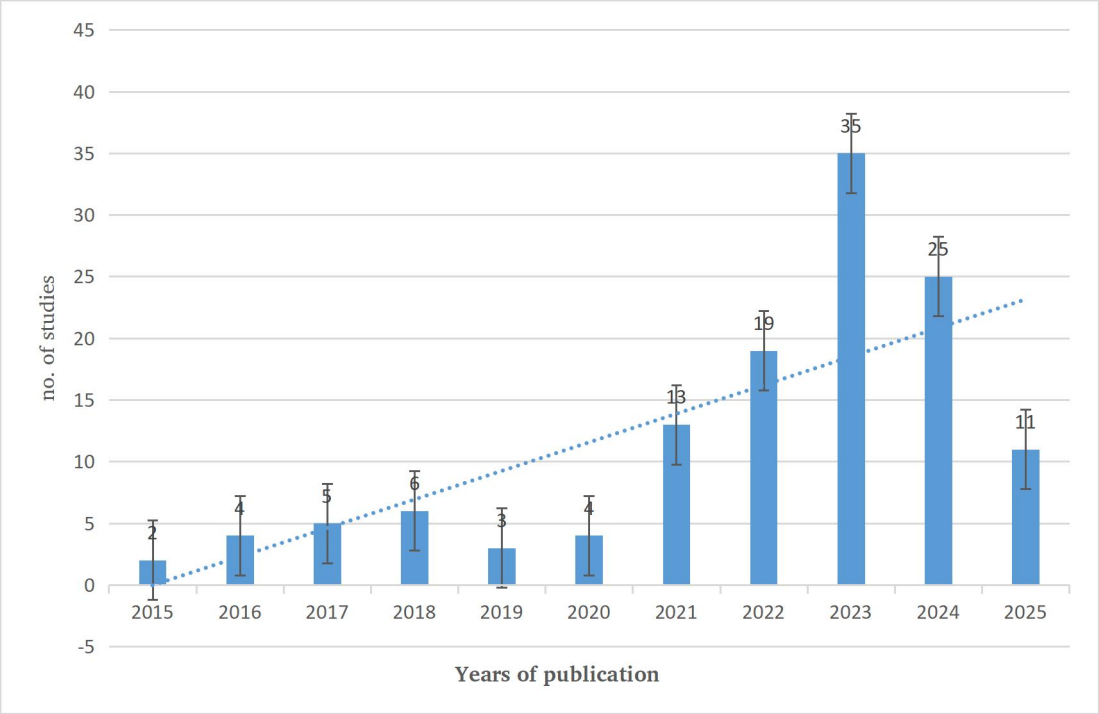


Figure 2 Year-wise distribution of the selected studies.

A specific implementation of the bagging algorithm named bagging ensemble classifier (BEC) and t-distributed stochastic neighbour embedding (t-SNE), was applied in the work of Somasundaram and Alli [57]. BEC was used to create a robust ensemble model that reduces overfitting, whereas t-SNE is a dimensionality reduction technique used to visualize high-dimensional data by projecting it into a low-dimensional space. Emerson et al. used various algorithms like batch-constrained Q-learning (BCQ), conservative Q-Learning, twin delayed deep deterministic policy gradient with behaviour cloning (TD3-BC), offline reinforcement learning (RL), etc. [60]. BCQ is a reinforcement learning technique designed for offline learning, and it uses a confined action space to reduce divergence and enhance decision-making in static datasets. CQL is also a reinforcement learning algorithm that is designed to enhance robustness in static datasets and penalize overestimation of out-of-distribution actions. TD3-BC algorithm combines reinforcement learning and supervised learning so that the offline policy optimization can be enhanced. Some researchers considered bootstrap resampling, which is a statistical technique used to generate multiple samples from the original dataset with replacement, resulting in robust estimation of model variability and accuracy in small and unbalanced datasets [45, 72]. Hahn et al. applied the bootstrapped cross-validation method, which combined the bootstrap resampling with conventional cross-validation to evaluate the performance of the model [73]. An advanced decision tree algorithm named C4.5 decision tree was used by Faruque et al. [74]. This algorithm builds upon ID3 by incorporating continuous attributes, pruning strategies, and improved handling of missing values. A non-parametric decision tree algorithm, i.e., classification and regression tree (CART), has also been explored in the literature, which supports classification and regression by partitioning the data into homogeneous groups [25, 41]. In clustering analysis, an advanced statistical approach called the cluster variation method was used to improve the interpretability of high-dimensional data through group formation by including variation within the clusters [75, 76]. For feature extraction purposes, researchers used a convolutional neural network (CNN) [22, 28, 63, 64]. A hybrid model that combined CNNs and long short-term memory (LSTM) is proposed by Chee et al. to perform the spatiotemporal data modelling. The CNN was used for feature extraction, while the LSTM for sequence learning [64]. To improve the classification accuracy, Nagaraj and Deepalakshmi used a hybrid approach that integrates enhanced support vector machines and deep neural networks (DNNs) [77]. A correlation method was used in predictive modelling to perform the multivariate analysis and feature selection [75]. This statistical technique evaluated the strength and direction of relationships between variables. To improve predictive performance in structured data, Nguyen et al. used crossed features and DNNs that combine engineered feature interactions with the representational power of Deep DNNs [78]. Similarly, Al-Qerem et al. employed the synthetic minority oversampling technique (SMOTE) to address the class imbalance and enhance the model performance on underrepresented categories [27]. Many authors used the SMOTE to balance the class distributions, improving the classification performance on imbalanced datasets [71, 78, 79]. Feng et al. used a technique formed by combining SMOTE and edited nearest neighbour (ENN) to clean noisy samples to form a high-quality and balanced dataset [62]. ENN is a data cleaning algorithm that is used to refine the datasets by removing misclassified instances. Additional machine learning models include decision tree classifiers, which recursively partition data based on feature values to support interpretable classification tasks [22, 28, 30, 32, 34]. A generative DL model named deep belief network was explored for learning the hierarchical representation of complex data [11]. Anusuya et al. applied deep convolution neural networks (DCNN), which are advanced convolutional architectures [22]. DCNNs are designed to advance complex visual recognition tasks. Naz et al. used an enhancement of the fuzzy clustering algorithm named modified fuzzy clustering method that improves the identification of cluster memberships for more precise partitioning [59].

Adhikari et al. implemented DenseNet, MobileNet, and VGG-19 methods [68]. DenseNet method that connects every layer to every other layer in a feedforward manner. MobileNet is a lightweight CNN. It reduces the computational cost through the depth-wise separation of convolutions. VGG-19 provides enhanced feature learning for complex image datasets. Fadhel et al. proposed a DL model named DFUFNet that was developed to analyse medical images so that medical anomalies could be easily detected and classified [80]. For high-dimensional datasets, the balance between feature selection and regularization is performed by using the combined properties of LASSO and Ridge regression. This combined approach is called elastic net (EN) [43]. Lu et al. used a technique named ensemble soft voting

classifiers to enhance the accuracy and stability [70]. A time-series analysis technique named exponentially weighted moving average was applied by Mathew et al. [12]. This technique is ideal for trend detection. Various researchers used an efficient gradient-boosted decision tree algorithm termed extreme gradient boosting (XGBoost) [14, 32, 35, 44, 81]. Wittler et al. used a neural network for basic pattern recognition tasks [28]. To evaluate the relationship between dependent and independent variables, Hamza et al. applied a statistical regression model based on the F-distribution, referred to as the F-Regressor [82]. A probabilistic classifier based on Bayes' theorem, named Naïve Bayes algorithm, has also been adopted in various studies [23, 33, 40]. This method provides robust performance even with small datasets. A gradient-based optimization method named stochastic gradient descent, and a Gaussian Naïve Bayes, i.e., a probabilistic classifier, has been incorporated by Nematollahi et al. [44]. For the feature selection and to improve the model accuracy, Li et al. used a hybrid method, GA XGBT, by combining genetic algorithm with extreme gradient boosting (XGBT) [83]. Ou et al. used gradient boosted decision trees to construct an ensemble of decision trees iteratively [84]. Another approach involves the general regression neural network (GRNN), a memory-based network that utilizes kernel density estimation for regression and classification tasks [23, 40]. Abnoosian et al. used Gaussian Naïve Bayes, especially for datasets with Gaussian-distributed features [85]. An ensemble technique named gradient boosted trees has also been employed by Allen et al. to minimize the prediction error [39], while some researchers used gradient boosting machine for this purpose [25, 43]. Furthermore, A statistical model known as the hidden Markov model has been applied as a statistical method to leverage latent states in the predictions [86].

Bilal et al. proposed a Hybrid CNN-SVD Model that is a combination of CNNs and Singular Value Decomposition (SVD), to enhance the feature extraction and minimize the dimensionality, and in contrast, the U-Net architecture has been applied for pixel-level prediction tasks, particularly in medical image segmentation [61]. To enhance predictive performance, Seth and Agarwal used a Hybrid DL Approach that integrates multiple DL techniques [87]. A hybrid model formed by combining CNNs and Shapley Additive exPlanations (SHAP) was also used to enhance the predictive power and interpretability [88]. CNN was used for feature extraction, whereas SHAP was used for explainability. Many researchers also used instance-based learning with K, which is a lazy learning algorithm and uses the K-nearest neighbours (KNN) approach [18, 75, 89]. A hybrid architecture, InceptionResNet Network, that is formed by combining Inception module's efficiency with residual connections, is used to enable deeper networks without gradient vanishing [65]. Sadeghi et al. implemented an isolation forest method, which detects anomalies by randomly partitioning the data using a tree-based structure [58]. Mishra et al. used a rule-based learner named JRip, which generated a set of If-Then rules from datasets [75]. It provides interpretable and effective classification for structured data. Many researchers used KNN as a simple but very powerful classification and regression method. It predicts the outcome based on the majority class. A partition-based clustering algorithm named K-Means Clustering method is used to group data points into K clusters [56, 59]. The least absolute shrinkage and selection operator (LASSO) method efficiently performs feature selection and minimises overfitting [38, 43, 45]. This method adds L1 regularization to the linear regression models. A very popular supervised learning algorithm, support vector machine (SVM), is used by multiple studies [13–15, 22, 36]. It finds the optimal hyperplane to categorize the data points in feature space and support linear and nonlinear decision boundaries. The light gradient boosting machine has also been explored to achieve high accuracy and scalability through histogram-based learning [13, 84]. This method is a fast, distributed gradient-boosting framework and is designed for large datasets. Curia used local interpretable model-agnostic explanations (LIME) and support vector classifier [32]. LIME provides model-agnostic explanations, whereas SVC is a type of SVM and is designed for binary or multi-class classification. A Linear Regression model was used to predict the continuous outcome based on the linear relationship between the dependent and independent variables [29, 42]. The Linear SVM, which is a variant of SVM, is used for the classification [44, 87, 90]. This method is computationally efficient for linearly separable data. Masi et al. applied the logic learning machine, a supervised learning algorithm, to generate decision rules for data classification [91]. Similarly, Alghamdi et al. used logistic model tree (LMT) and Multiple Linear Regression, which is an extension of Linear Regression [30]. The LMT combines decision trees with logistic regression and is used to improve the classification in datasets with mixed data types. Logistic Regression is a probabilistic statistical model that is used for binary or multinomial classification tasks [13, 27, 30, 32, 33]. A recurrent neural network architecture named LSTM is used to capture long-term dependencies in sequential data [19, 60, 63, 64, 66]. It is widely used in time series and text analysis. Islam et al. and Theerthagiri et al. applied Multilayer Perceptron (MLP), a feedforward neural network class with multiple hidden layers [18, 92]. This method is adept at learning complex nonlinear mappings in the data. Multivariate logistic regression is a statistical method used to model relationships between multiple predictor variables and a binary or multinomial outcome [41, 45]. Xu and Hu used a variant of CNN named One-Dimensional Convolutional Neural Network (1DCNN), designed for sequential or temporal data that leverages 1D convolutions for extracting local patterns effectively [33]. Recent studies used Random Forest to construct multiple decision trees and aggregate their outputs to improve accuracy and prevent overfitting [27–31]. Peng et al. proposed a feature selection technique known as the Random-Forest-Based Boruta Algorithm, which identifies relevant variables by comparing them with randomized shadow features [93]. A type of artificial neural network named Radial Basis Function (RBF) has been explored for interpolation and classification tasks [89]. To manage the long-term dependencies in time-series data, Sharma et al. proposed a hybrid architecture of sequential processing power of RNNs and the memory capabilities of LSTM, named recurrent neural network - long short-term memory (RNN-LSTM) [66]. An approach named recursive feature elimination (RFE) is used for the iterative removal of the less important features based on coefficients, and to optimize the input dataset for predictive modelling [71]. Ryu et al. implemented two methods, ResNet101 Model, and You Only Look Once V3 Algorithm [94]. ResNet101 model is a deep residual network with 101 layers and is suitable for image and video recognition tasks, whereas it is a real-time object detection algorithm that balances accuracy and speed. Shahin et al. applied restricted boltzmann machines, which are energy-based generative neural networks. It learns representations through the optimization of the reconstruction of input data [11]. Shaheen et al. proposed two deep learning models, LeDNet and HiDenNet, for early diabetes prediction [69]. These models outperform the existing models in stability and accuracy. Nath et al. used stacking regression, a meta-learning approach for enhancing overall predictive accuracy [29]. For specific applications requiring precise accuracy, a specific DL framework named TFDA-1 was introduced by Li et al [83]. Al-Absi et al. used a simplified version of the VGGNet series with 11 layers [67]. It provides a balance of depth and computational efficiency for image recognition tasks. An ensemble learning model designed by combining Logistic Regression and Random Forests using weighted voting for the optimization of classification performances was proposed by Fazakis et al. and named as Weighted Voting LRRFs Model [38]. In addition, an advanced gradient-boosted framework named XGBoost has been widely applied in recent studies [14, 32, 35, 37, 44]. It is used for regularization and reduction of overfitting and supports parallel processing. All the techniques are presented in Supplementary Table S3.

RQ-1 (i): Comparative performance analysis of AI models in diabetes prediction and diagnosis

To provide practical insights, we compared major AI methods across data types and clinical settings highlighted in the literature (see Supplementary Table S4). Deep learning, especially CNNs,

DenseNet121, and ResNet50, dominates imaging tasks like diabetic retinopathy screening, with accuracy above 90% and AUC close to 0.95. When working with structured records from electronic health records (EHRs), ensemble techniques such as Random Forest and XGBoost shine, yielding AUCs between 0.87 and 0.94 thanks to their robustness against mixed features and class imbalance. LSTM and GRU models successfully track long-term trends for time-series data, proving useful in glucose monitoring and complication forecasts. Although simpler algorithms such as Naïve Bayes and KNN are easy to set up, they struggle with the complexity found in many modern datasets. Finally, hybrid and explainable approaches, for example, a CNN coupled with SHAP values or Elastic Net, strike an appealing balance, offering strong predictions while remaining interpretable, a crucial step for real-world clinical rollout through this synthesis. It is noted that the choice of models should be motivated by the data characteristics, the requirement of model interpretability, and the clinical setting. It should also guarantee high accuracy and trustworthiness of AI-based diabetes care.

RQ-2: What datasets are commonly used to train and validate the AI models for diabetes prediction?

In this review work, we classify the datasets into three categories based on their sizes. The datasets with hundreds to a few thousand samples are considered as small datasets, and the datasets with tens of thousands to hundreds of thousands are considered as medium-sized, whereas datasets with millions or more samples are considered as large datasets. In this study, most of the datasets used are of small size, as presented in Table 2.

RQ-2 (i): Dataset limitations and generalizability

One big takeaway from our review is that almost every study leans on small, nearly identical datasets, which limits how far its AI models can travel when we try to use them worldwide. A lot of papers still run with classics like the PIMA Indian Diabetes or Sylhet Dataset; both are bite-sized and usually gather data from only one community or region. Because of this, a model may shine during training but then struggle once it meets a broader or more varied group. Adding to the problem, these sets stick to neat, row-and-column numbers and miss out on behaviors, income levels, and even genetic info that matter to a full picture of diabetes risk. So, as researchers push their algorithms into clinics, that low diversity raises the odds of skewed predictions and unfair treatment. If we want fairer, smarter tools, the field really needs to start building and sharing big, mixed, and multimodal datasets in the open. Approaches like federated learning and partnerships across hospitals can help keep patient privacy intact. Table 3 concisely describes major datasets found in the literature and their characteristics and applications to underline the need to work on such limitations to have a global application of such data and their analysis.

RQ-3: Which data pre-processing techniques are most effective in enhancing the performance of AI models used for diabetes prediction?

Based on our review, we found various data pre-processing techniques used by the researchers, as shown in Supplementary Table S5. A statistical method named bootstrap resampling was used to generate multiple datasets [45, 72, 73]. Hahn et al. used a variation of this technique, named bootstrapped cross-validation, to create multiple training and validation sets to improve the model evaluation [73]. Mishra et al. used the correlation method to measure the relationship between the different features in a dataset, so the redundancy can be reduced by removing features with a high correlation [75]. For the data augmentation, the SMOTE was used to address class imbalance in datasets [71, 78, 79]. For feature selection, the LASSO technique has been implemented, which shrinks the coefficients of less important variables toward zero [38, 43, 45]. McDaniel et al. used a regularization technique named Elastic Net that combines LASSO and Ridge Regression to enhance the feature selection [43]. Sadeghi et al.used the Isolation Forest method, isolating outliers based on decision trees [58]. Guleria et al. employed a feature selection method MRMR that selects the most relevant features while minimizing redundancy [26]. Alhassan et al. used a dimensionality reduction technique named PCA that transforms correlated features into a smaller set of uncorrelated principal components [19]. Another feature selection technique named RFE is used by Jasim et al., which recursively removes the least important features and provides a ranking based on their contribution [71]. Feng et al. employed a hybrid technique named SMOTEENN that is formed by combining the SMOTE and ENN [62]. Somasundaram and Alli used a nonlinear dimensionality reduction technique named t-SNE for visualizing the high-dimensional data [57], as illustrated in Table 4.

Table 2 Various types of datasets are used in the literature

Dataset Name	Dataset Size	References
PIMA Indian Diabetes Dataset	Small	[13, 16, 18, 21–23, 25, 40, 42, 51, 62, 71, 77, 79, 95–105]
MIMIC-III Dataset	Medium	[28, 79]
Healthcare Diabetes Dataset	Small	[15, 18, 31]
Messidor Dataset	Small	[36, 61, 106, 107]
NIH NIDDK Central Repository	Large	[29]
EyePACS Dataset	Medium	[61, 87]
South Korea HER	Large	[39, 78]
Sylhet Diabetes Dataset	Small	[25, 79]
NHANES Dataset	Medium	[25]
KoGES health examinee (HEXA) study Dataset	Large	[73]
KoGES cardiovascular disease association study (CAVAS) Dataset	Large	[73]
Hamad Medical Corporation and Qatar Biobank.	Large	[67]
Taiwan Biobank	Large	[108]
UK Biobank Dataset	Large	[109]
DIARETDB0 / DIARETDB1 Dataset	Small	[36, 61]
Kaggle Diabetes Dataset	Small	[17, 20, 105, 110]
Iraq Patient Diabetes Dataset	Small	[85]
Shuguang Dataset	Small	[83]
e-ophtha Ex public Dataset	Medium	[106]
St John Hospital, Hong Kong.	Large	[65]
University of California, Irvine Diabetes Mellitus Dataset	Medium	[51]

Table 3 Summary of key datasets used in reviewed studies

Dataset Name	Size Category	Geographical Origin	Data Type	Primary Use in Studies
PIMA Indian Diabetes Dataset	Small	India (U.S.-based population)	Structured (tabular)	Binary classification, risk prediction
Sylhet Diabetes Dataset	Small	Bangladesh	Structured (tabular)	Diagnosis
MIMIC-III	Medium	USA	EHR, time-series	Risk prediction, longitudinal modeling
EyePACS Dataset	Medium	USA	Retinal images	Diabetic retinopathy detection
NHANES	Medium	USA	Structured (tabular)	Lifestyle-related risk prediction
UK Biobank	Large	United Kingdom	Genomic, clinical, imaging	Multimodal prediction and analysis
Taiwan Biobank	Large	Taiwan	Genomic, clinical	Population-wide risk stratification
Hamad Medical Corp/Qatar Biobank	Large	Qatar	Structured, genetic	Regional population health modeling
KoGES (HEXA & CAVAS)	Large	South Korea	EHR, structured data	Longitudinal risk prediction
Messidor Dataset	Small	France	Retinal images	Diabetic eye disease detection
DIARETDB0 / DIARETDB1	Small	Finland	Retinal images	Diabetic retinopathy classification

Table 4 Data pre-processing techniques used in the literature

Method	References
Bootstrap resampling	[45, 72]
Bootstrapped cross-validation	[73]
Correlation method	[75]
SMOTE	[27]
Elastic net method	[43]
Isolation forest method	[58]
LASSO method	[38, 43, 45]
MRMR	[26]
RFE	[71]
SMOTEENN hybrid sampling technique	[62]
t-SNE	[57]
PCA	[19]

LASSO, least absolute shrinkage and selection operator; MRMR, minimum redundancy maximum relevance; PCA, principal component analysis; RFE, recursive feature elimination; SMOTE, synthetic minority oversampling technique.

RQ-3 (i): Analytical insights into preprocessing techniques for diabetes prediction

RQ-3 now digs into how each preprocessing step moves the needle on diabetes-prediction accuracy to answer the reviewer's question better. It doesn't just list methods; it ties each one to a real data issue: class imbalance, high dimensionality, noise, or feature overlap. Take SMOTE and its SMOTE-ENN blend: by creating synthetic cases, they lift recall and a AUC numbers in lopsided datasets. LASSO, Elastic Net, MRMR, and RFE trim away redundant signals, cutting overfitting while speeding up training and keeping results understandable. When features explode, PCA or t-SNE brings the load down, making models run faster and crash less often. Isolation Forest prunes rogue outliers from clinical notes, and Bootstrap Resampling along with Bootstrapped Cross Validation steadies' performance checks even when data is scarce or messy. Every tweak-leveraging time, memory, or signal quality-shows up in cleaner accuracy charts, stress-testing these steps as core pieces of a robust diabetes-AI pipeline.

RQ-4: What are the key limitations of the present AI-based diabetes prediction methods, and how might these limitations be addressed?

Although AI-based techniques used for diabetes prediction have shown great potential, several challenges still limit the accuracy and practicality of these techniques in the healthcare domain. One of the major challenges is that the traditional ML models are not able to completely understand the complexity of diabetes, which involves various factors such as blood pressure, blood sugar, lifestyle habits, etc. [27]. The lack of research on how blood sugar levels fluctuate over time is another issue because the real-time monitoring becomes less reliable [100]. Another issue is that many models focus only on type-2 diabetes and ignore type-1 and gestational diabetes, minimizing their applicability to many patients [89]. The existing models frequently exclude important factors like lifestyle, diet, and environmental influences, resulting in incomplete predictions [77, 95]. Many researchers also failed to integrate genetic and omics data, which could increase the prediction accuracy by identifying high-risk individuals based on their DNA [77].

Another major issue is data-related challenges. Many AI models face the problems of noise and data redundancy, which can produce inaccurate classification and misleading results [89]. Unbalanced datasets are also a concern because models trained on biased data may not accurately predict diabetes in underrepresented categories of patients, like people with prediabetes [29]. High computational requirements are also an issue because DL models sometimes require advanced hardware, which makes them challenging to use in clinics with limited funding [61]. Furthermore, certain AI-based systems depend on sluggish internet connections, which can cause delays in real-time analysis and decision-making in remote areas [102]. Another significant limitation is that the decision-making process of most AI models is so difficult for doctors and patients to understand that it causes a loss of trust in AI predictions, and the clinical implementation becomes challenging [111]. Some models only focus on the diagnosis of diabetes and do not provide predictions of the related complications, like retinopathy, diabetic neuropathy, etc. [31]. Furthermore, longitudinal studies that monitor the progression of

diabetes over time are not present in the existing research, which limits the capacity of the AI models to provide health predictions for a long time [100].

These issues can be addressed by developing more inclusive models that can consider various types of diabetes using real-time monitoring of glucose, so that the early detection of diabetes can be improved. For AI models to be accessible in the diverse healthcare domains, they should also be optimized for minimum computational power. Lastly, a more comprehensive approach to managing diabetes can be provided by expanding the AI models for the prediction of the various complications related to diabetes and conducting long-term research. By addressing these issues, the accuracy, reliability, and use of AI-based diabetes predictions in healthcare can be improved.

Discussion

In recent years, researchers have built increasingly sophisticated AI algorithms that predict and diagnose diabetes with impressive accuracy, yet few of these proof-of-concept systems have moved into everyday clinics. A major roadblock stems from the lengthy regulatory and validation process that any new medical tool must clear. Most algorithms get tested on historic patient records in silico, yet few have been examined in live healthcare settings over extended periods [112]. Therefore, before granting clearance, agencies such as the FDA and EMA insist on solid, real-world proof of safety, effectiveness, and consistent performance across diverse populations [113]. Because harmonized protocols and detailed, transparent criteria for clinical-grade AI are still taking shape, many promising models sit on the shelf rather than at the bedside [114]. Interpretability presents another concern, particularly for deep networks such as convolutional and recurrent layers that power many diabetes pipelines today [115]. When clinicians see these systems described as black boxes, mistrust naturally follows, and professionals are unwilling to act on recommendations that they cannot fully understand. Efforts to tame this opacity through tools like SHAP and LIME show promise, yet embedding these insights into routine workflows and making them timely, relevant, and clinically meaningful remains superficial at best [116].

Doctors need predictions they can trust, and they also want a clear story about how an AI concluded, especially when lives are on the line, as with early diabetes screening or managing serious complications. That expectation points to an urgent demand for interpretable systems that lay out step-by-step reasoning, preferably by blending rules-based logic with what statistics has learned from data [117]. User interfaces and integration hurdles still slow the spread of AI in hospitals around the world. Many models are built on dazzling math yet ignore how busy clinicians actually work or how well they plug into existing hospital programs [118]. Because of that, even technically sound tools can collect dust, squeezed out by awkward screens or a workflow that pulls staff in the opposite direction. Meaningful adoption, therefore, calls for systems slotted right into Electronic Health Records, designed for quick clicks, and able to spit out clear, timely alerts. On top of that, uneven Internet access, different levels of tech training, and budget limitations across regions keep some facilities from ever seeing the same gains.

Federated learning and other privacy-first AI methods let hospitals and research centres train models together without ever moving raw patient records, and early tests show the idea works [119]. Take Merck's Swarm Learning initiative: The system still predicts disease risk by keeping data local on each edge device while respecting institutional ownership. Similar pilots that tap the MIMIC-III ward database and the EyePACS diabetic-retinopathy images have added noise through differential privacy, proving models can run under tight confidentiality budgets [120]. These projects prove the tech can be built, but moving from small studies to everyday care demands close teamwork among coders, doctors, network engineers and health policymakers. Success will depend on systems that explain their decisions, meet evolving rules, offer clear, user-friendly screens, and follow fair governance, especially in clinics that serve diverse or under-resourced communities.

Table 5 provides an overview of the greatest obstacles to implementing AI in diabetes screening in the health care contexts and identifies possible ways of overcoming them. The major obstacles refer to regulatory barriers because of the absence of standardised validation due to the lack of a long-lasting deep learning model, also known as the black box issue, which complicates clinical confidence in a deep learning model, as well as a weak interface with the current EHR systems [121]. Moreover, the issue of data privacy restricts cooperation, reduces data diversity, and the use of small, homogeneous datasets influences the generalizability of AI models. The proposed solutions include the cultivation of standardised validation guidelines, the integration of explainable artificial intelligence (XAI), the clinician-centred design of user interface, the application of federated learning to secure data sharing chores, and the practice of multi-site/diversified data usage.

Table 5 Barriers and enablers for clinical deployment of AI in diabetes screening

Category	Barrier	Impact	Example/Pilot Project	Potential Solution
Regulatory	Lack of standardized validation and slow approval processes	Delays clinical adoption; lack of trust from regulators	FDA's limited AI approvals in the diabetes domain	Develop harmonized AI validation protocols and clear regulatory frameworks
Interpretability	Deep learning models often function as "black boxes"	Clinicians struggle to trust or understand AI decisions	Use of SHAP/LIME is not yet widely integrated into clinical tools	Incorporate explainable AI (XAI) and hybrid interpretable models
User Interface & Workflow	Poor EHR integration and non-intuitive interfaces	Interrupts clinical workflow; leads to underuse or rejection of tools	Many models trained on MIMIC-III remain research-bound	Co-design with clinicians; seamless EHR integration; usability testing
Data Privacy	Concerns over patient data sharing and breaches	Limits cross-institutional collaboration; reduces data diversity	Swarm Learning (Merck), federated learning on EyePACS	Use federated learning and differential privacy to enable secure, decentralized AI
Generalizability	Reliance on small, homogenous datasets	Models fail on underrepresented populations; risks biased decisions	The PIMA Indian dataset is used in most studies	Encourage data sharing, use of multi-site, multi-ethnic datasets

Conclusion

This SLR compiles 127 studies and maps how artificial-intelligence techniques assist in predicting and diagnosing diabetes. Evidence shows that clinicians and researchers favour supervised models-Random Forest, SVM and XGBoost-when working with structured health records, because these algorithms yield strong, easy-to-explain outputs. CNN and LSTM networks consistently achieve high sensitivity and specificity for image-rich tasks such as retinopathy screening. Yet even the best feature-selection methods, including RFE and SMOTE, yield only incremental gains, partly because most studies draw on small, homogenous cohorts that skew age, gender, and ethnicity.

To move artificial intelligence from labs to everyday clinics, we outline a five-stage roadmap: (1) design algorithms that clinical teams can interpret; (2) validate those models on diverse, up-to-date patient datasets; (3) secure regulatory and ethics approval while documenting transparency and fairness; (4) deploy tools through seamless links with hospital software and wearable sensors; and (5) monitor performance continuously after rollout to protect patient safety. Long-lasting adoption will also depend on open-access data banks, routine feedback between clinicians and coders, federated-learning frameworks that respect privacy, and ambitious, multi-year trials that measure both clinical and economic outcomes. These goals and actions are meant to achieve the intersection of AI innovation with its constructive and ethical application to diabetes care on a global scale.

Data availability

No datasets were generated or analysed during the current study.

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